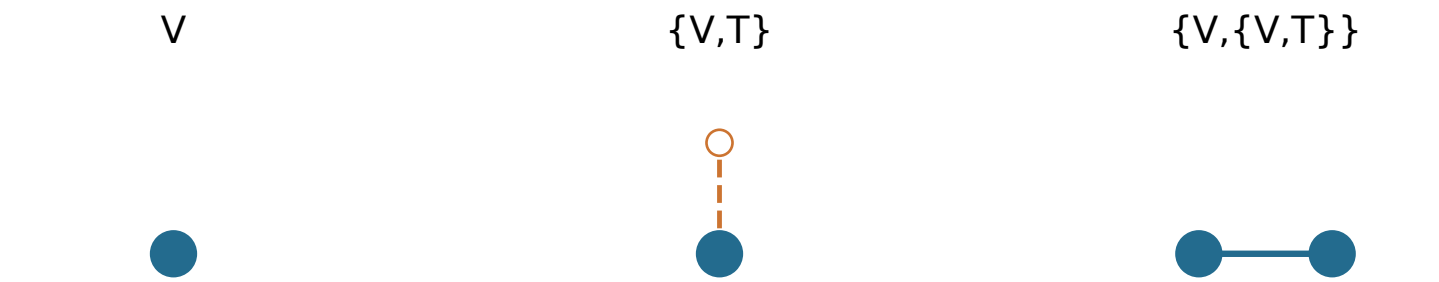Tree contractions for V(q) = q²/2 + q⁴/4